Kunal Anand

Backend Engineering | Systems | Applied AI
+91-9473248971 | Email | LinkedIn | GitHub | Portfolio

Professional Summary

End-to-end software development focused on backend systems and applied AI, with hands-on experience building APIs, multi-tenant applications, secure workflows, and AI-powered retrieval systems.

Education

R.R.S.D.C.E., Begusarai (Bihar Engineering University) 2023 – 2027
B.Tech, Computer Science and Engineering (Data Science) *Begusarai, India*

Technical Skills

Languages: Python, JavaScript, C++, SQL
Backend & Frameworks: FastAPI, Node.js, Express.js, Next.js, REST APIs
Databases: PostgreSQL, MongoDB, MySQL, Neo4j
AI / Cloud: Gemini API, LangChain, RAG, GraphRAG, Hugging Face, BigQuery, Cloud Storage, Cloud Run
Backend Concepts: Authentication, JWT, Rate Limiting, API Design, Input Validation
Tools: Git/GitHub, Docker, Linux, Postman

Experience

FlyRank AI | **GitHub** Jul 2026 – Sep 2026
Backend Engineering Intern *Remote*
- Built a multi-tenant lead-capture backend from scratch using FastAPI, PostgreSQL, SQLAlchemy, and Alembic, with tenant-scoped APIs and configurable embeddable widgets.
- Implemented origin validation, rate limiting, honeypot-based spam protection, payload-size limits, and idempotent submission handling to protect public-facing endpoints.
- Designed background notification processing using a transactional outbox, retry handling, and failure recovery; built tenant-scoped dashboard APIs and automated tests.

Startly Innovations Pvt. Ltd. Apr 2026 – May 2026
Backend Intern *Remote*
- Built backend APIs using Node.js, Express.js, and MongoDB for a role-based registration platform supporting Founder and Audience workflows.
- Designed admin APIs for participant filtering, search, deletion, and structured data export; implemented IP-based rate limiting, validation, sanitization, and duplicate-registration prevention for public-facing APIs.

Projects

ObserverModel | *AI Runtime Observability Platform* | GitHub | Article 2026
- Built an AI-agent observability platform that parses execution logs from Claude Code to detect retry loops, repeated reads, excessive file enumeration, and security-sensitive behavior.
- Implemented deterministic anomaly detectors and integrated Gemini-powered root-cause analysis, with human approval before recommendations are surfaced.
- Developed token-cost attribution and historical execution analytics using BigQuery and Cloud Storage, and deployed the FastAPI application on Google Cloud Run.

Multilingual RAG Support Platform | *Team Lead* | *SIH 2025 Grand Finalist* | GitHub Dec 2025
- Implemented a Neo4j-based GraphRAG retrieval pipeline to incorporate relationships between knowledge entities for more context-aware answers.
- Integrated the GraphRAG retrieval workflow into the chatbot's backend to support knowledge-grounded responses.
- Led a six-member team building workspace-specific knowledge bases, embeddable chat widgets, and confidence-based escalation to human agents.

Secure Gemini Journal | *Secure AI Application* | GitHub | Article 2026
- Built a multi-user AI journaling application using Next.js, Firebase Authentication, Firestore, Cloud Run, and Gemini, with backend-enforced conversation ownership and user data isolation.
- Implemented a selective response pipeline with SILENCE, ACK, and REFLECT outcomes, invoking Gemini and retrieving historical context only when reflection was needed.
- Designed safety checks and server-authoritative message persistence, addressing unauthorized access, prompt injection, context poisoning, and API abuse.